

# Programmable Interrupt Controller - IP Card

Structured design summary

| Field                                 | Value                             |
|---------------------------------------|-----------------------------------|
| Basic Information                     |                                   |
| Licensing                             | Open-source (CERN-OHL-S-2.0)      |
| Target Fpga Or Asic                   | FPGA                              |
| Design Language                       | SystemVerilog                     |
| System-Level Features                 |                                   |
| Coupling Type                         | Loosely-coupled                   |
| Data Types                            | N/A                               |
| Isa                                   | RISC-V + memory-mapped interrupts |
| Peak Ops Per Cycle                    | N/A                               |
| Interfaces                            | AHB Lite                          |
| Supported Operations                  | N/A                               |
| Compiler Support                      | Standard Memory Instructions      |
| External Memory Required              | No                                |
| Cache Or Scratchpad                   | N/A                               |
| Architecture                          |                                   |
| Type                                  | N/A                               |
| Num Clock Domains                     | 1                                 |
| Control Interface > Isa Extension     | No                                |
| Control Interface > Memory Mapped     | Yes                               |
| Control Interface > Protocol          | AHB Lite                          |
| Memory Interface > Interface Protocol | AHB Lite                          |
| Memory Interface > Hierarchy Level    | ?                                 |

| Field                                                   | Value                               |
|---------------------------------------------------------|-------------------------------------|
| Memory Interface > Cached                               | No                                  |
| Memory Interface > Iommu                                | No                                  |
| Interrupts                                              | Async Interrupts                    |
| <b>Microarchitecture</b>                                |                                     |
| Parametrization > Parametric Num Units                  | Generated by parametrized Generator |
| Parametrization > Parametric Config                     | No                                  |
| Programmability > Programmable Cores                    | No                                  |
| Programmability > Isa Extension                         | N/A                                 |
| <b>Software</b>                                         |                                     |
| Compiler > Requires Specialized Compiler                | No                                  |
| Compiler > Supported Compilers                          | ?                                   |
| Hardware Abstraction Layer > Hal Functions              | Register definition header          |
| High Level Api > Has High Level Api                     | No                                  |
| High Level Api > Sdk Repository                         | No                                  |
| High Level Api > Has Domain Specific Compiler           | No                                  |
| <b>Integration</b>                                      |                                     |
| Ip Distribution > Manifest Type                         | Bender.yml                          |
| Ip Distribution > Standalone Simulation                 | Formal Verification                 |
| Ip Distribution > Standalone Simulation Sw Requirements | Formal Tools                        |
| Ip Distribution > Integration Docs Or Examples          | Yes                                 |
| Ip Distribution > System Rdl                            | No                                  |
| Synthesis > Is Synthesizable                            | Yes                                 |
| Synthesis > Fpga Synthesis Examples                     | No                                  |
| Synthesis > Asic Synthesis Examples                     | No                                  |

| Field                                    | Value |
|------------------------------------------|-------|
| Simulation > Closed Source Simulation    | No    |
| Simulation > Open Source Simulation      | No    |
| Evaluation > Ppa Results Available       | No    |
| Physical Implementation                  |       |
| Asic Area > Num Gates                    | N/A   |
| Asic Area > On Chip Internal Memory Size | N/A   |
| Asic Area > Target Clock Frequency       | N/A)  |